

Study Guide

Every concept in this project, explained, in the order it appears. Built so you can defend any sentence of the report in an interview.

How to use this

Read it once end to end. Then use the question list at the back as a self-test. If you can answer those without notes, you understand the project. Anything you can't answer, come back to the section it came from.

Part 1 — The physics you are measuring

1.1 dBm, and why the numbers are negative

dBm is power expressed on a logarithmic scale relative to one milliwatt. 0 dBm is 1 mW. WiFi signals arriving at a client are tiny fractions of a milliwatt, so the numbers are negative — -50 dBm is 0.00001 mW.

The logarithm is what matters in practice: **every 3 dB is a doubling or halving of power, every 10 dB is a factor of ten.** So -60 is not 'slightly worse' than -50 — it is ten times less power. This is why a 6 dB difference across a wall is significant and why you cannot average dBm values naively.

Reading	Verdict	Meaning
-30 to -45	Excellent	Essentially under the AP. Top modulation rates.
-46 to -60	Very good	Comfortable for anything including video.
-61 to -67	Good	-67 is the standard design floor for voice and video.
-68 to -72	Marginal	Rates drop hard, roaming gets sticky.
-73 to -80	Poor	This is what a user calls a dead zone.
Below -80	Unusable	Client drops or refuses to associate.

Your survey: mean best 5 GHz signal was -49.7 dBm on the ground floor. Every point beat -67. That is why 'no coverage problem' is a defensible claim.

1.2 Noise floor and SNR

The **noise floor** is background RF energy with no WiFi in it, typically -90 to -95 dBm indoors. **SNR** is signal minus noise, expressed in dB: a signal of -65 against noise of -92 gives 27 dB SNR.

SNR predicts link quality better than raw signal does. Strong signal with poor SNR is still a broken link — and it points at interference rather than distance, which is a completely different fix.

Your survey: the Android build reported no noise data at all. That is why there is no SNR finding anywhere in your report, and why you say so explicitly. Knowing what you could not measure is part of the analysis.

1.3 Attenuation

Signal loss passing through material, measured in dB. Approximate values at 5 GHz:

Material	Loss	Material	Loss
Gypsum board partition	3–5 dB	Solid wood door	5–10 dB
Plain interior glass	3–8 dB	Brick wall	8–15 dB
Low-E coated glass	15–30 dB	Concrete / CMU	15–25 dB

Material	Loss	Material	Loss
Loaded bookshelves	5–15 dB	Metal shelving	15–30 dB
Human body	3–6 dB	Elevator shaft	30 dB+

Two things this explains in your building. The brick-cavity-over-CMU wings are near opaque at 5 GHz, so those volumes must be served from inside them — which partly explains the AP count. And the Guardian SN 68 glazing is low-emissivity: its metallic coating attenuates far more than plain glass, so that big glass reading room leaks much less signal outdoors than it looks like it should. That is a genuinely counterintuitive point worth being able to make.

It also explains why holding the phone away from your body mattered. Your torso is 3–6 dB of attenuation; inconsistent posture would have been indistinguishable from real signal variation.

Part 2 — How WiFi is organised

2.1 SSID, BSSID, and why counting matters

Term	What it is
SSID	The network name humans see. Broadcast from many APs at once. Counting SSIDs tells you nothing about how many APs exist.
BSSID	The MAC address of one radio serving one SSID. This is the unique identifier you track.
Radio	One physical transmitter, on one band. An AP typically has two (2.4 and 5 GHz).
Access point	The physical box. Two radios, each advertising several BSSIDs — one per SSID it serves.

So one AP serving three SSIDs on two bands produces six BSSIDs. Your survey saw 94 distinct BSSIDs. That was never 94 access points.

The technique that solved it. The library's BSSIDs share a structured prefix: 8a:15:04:XX:XX:Yn for 2.4 GHz radios and 8a:15:14:XX:XX:Yn for 5 GHz. Grouping on the middle bytes collapsed 94 BSSIDs to **eight access points**. The leading 8a has the locally-administered bit set, which is typical of controller-managed enterprise gear issuing virtual BSSIDs. This is a transferable trick — learn it.

2.2 Channels and the 2.4 GHz problem

The 2.4 GHz band is divided into channels 1–11 in the US, but each channel is wider than the spacing between them. Only **1, 6 and 11** do not overlap. Everything else partially overlaps its neighbours.

This creates two distinct problems, and the difference matters:

Type	What happens	Severity
Co-channel interference (CCI)	Two APs on the <i>same</i> channel. They can decode each other, so they politely take turns — carrier sense makes them wait. Not corruption; contention. Everyone gets slower as clients multiply.	Bad, and the commonest real problem
Adjacent-channel interference (ACI)	APs on <i>partially overlapping</i> channels. They cannot decode each other, so they transmit over each other. Genuine corruption and retransmission.	Worse than CCI

Your F-01: five of eight APs on channel 6, with 24 distinct BSSIDs audible there against seven on channel 11. That is co-channel contention, self-inflicted, fixable for free.

2.3 5 GHz, channel width, and the trade

5 GHz has far more spectrum, divided into three ranges: **UNII-1** (36–48), **UNII-2** (52–144, shared with weather radar and therefore subject to DFS), and **UNII-3** (149–165).

DFS — Dynamic Frequency Selection — means an AP on those channels must vacate if it detects radar. Some vendors and some clients avoid DFS channels for that reason, which is why UNII-3 gets crowded while a large clean range sits empty.

Channel width is the central trade in WLAN design. A wider channel carries more bits per transmission but consumes more spectrum, leaving fewer non-overlapping channels for your other APs:

Width	Non-overlapping channels available	Best for
20 MHz	Most	Very dense deployments, high client counts
40 MHz	Half as many as 20	Usually the right answer in a multi-AP building
80 MHz	Few	Homes and low-density spaces

Width	Non-overlapping channels available	Best for
160 MHz	Almost none	Single-AP, one-client scenarios

Your F-02, and the arithmetic behind it. An 80 MHz channel with primary 149 occupies 149 through 161 — it swallows 153 and 157 entirely. The library has three APs on 149, two on 153, one on 157. Six of eight radios are stacked inside one overlapping block while UNII-1 and UNII-2 sit almost unused. Being able to do that spectrum arithmetic out loud is what makes the finding credible.

Part 3 — Design concepts

3.1 Coverage versus capacity

Coverage asks: can a client hear an AP well enough to work? **Capacity** asks: can the network serve everyone who is present at once?

They call for opposite responses. A coverage problem is solved by more or better-placed APs. A capacity problem is solved by more channels, narrower channels, or reduced power — and adding APs often makes it worse.

This is the single most important idea in your whole report. Your survey found perfect coverage and a capacity/contention problem. The naive recommendation — 'add access points' — would have made the actual problem worse. Getting that the right way round is what makes this look like real analysis rather than tool operation.

3.2 Cell overlap and roaming

A **cell** is the area one radio effectively serves. Neighbouring cells should overlap by roughly **15–20%** — enough that a walking client can move from one to the next without dropping, but not so much that it hears half the building.

Sticky client is the failure mode: a device clinging to a distant, weak AP because it still technically works, instead of roaming to a strong nearby one. Excessive overlap makes this more likely, because the client sees many similar candidates and has no clear reason to switch. Roaming is a client-side decision; the network can only influence it.

Your F-03 and F-05. Hearing 7.3 library radios at once — peaking at 13 — is far beyond 15–20% overlap. And F-05 showed the overlap is three-dimensional: six of eight APs reach both floors through the double-height reading room. Clients near that void see an unusually large candidate set.

3.3 Transmit power

Turning power *down* is a legitimate and frequently correct fix. It shrinks each cell, reduces how many APs a client can hear, cuts co-channel contention, and encourages cleaner roaming. It feels wrong the first time you hear it — less signal, better network — which is exactly why being able to explain it lands well.

The constraint: you can only reduce power while every location still meets the design floor. Your data shows a mean of -49.7 dBm against a -67 threshold. **That is roughly 17 dB of headroom**, which is a lot of room to turn down.

3.4 Survey types

Type	What it is	When
Passive	Listen to beacons. No association, no traffic. What you did.	Auditing an existing network
Active	Associate and measure real throughput, retries, roaming.	Troubleshooting performance complaints
Predictive / Planning	No visit. Simulate from floor plan, materials and virtual AP placement.	Designing before it exists; modelling a fix
AP-on-a-Stick	Temporarily mount a real AP, measure its true cell, repeat.	Validating a predictive design physically

Part 4 — Method and instrument

4.1 Why the two Android settings mattered

- **Precise location permission.** Android returns zero scan results to an app without it. Not a quality issue — a total-failure issue.
- **WiFi scan throttling.** Android 10+ caps scans at roughly four per two minutes to save battery. Left on, you walk an entire building collecting near-duplicate cached readings and never notice.

4.2 Your instrument's limits, and why you state them

Property	Yours	Consequence
Standard	Wi-Fi 5 (802.11ac)	Cannot associate at Wi-Fi 6 rates
Bands	2.4 and 5 GHz only	No 6 GHz visibility at all. Your biggest blind spot.
Streams	1×1	Caps at 433 Mbps on 80 MHz. The 433 figure is how you know it is 1×1: 80 MHz × 1 stream × MCS9 × short GI = 433.
Calibration	Consumer chipset	±3–6 dB absolute uncertainty. Relative comparisons within your own survey remain valid — which is where all your findings come from.

Say this unprompted in an interview. A candidate who volunteers their measurement limitations reads as someone who understands measurement. A candidate who doesn't reads as someone who doesn't know what they don't know. It is a rare and cheap way to look senior.

4.3 The map-coverage trap

NetSpot reports a 'map coverage %' that looks like a quality score. It is not. It measures **spatial spread** — how much of the floor plan falls within interpolation range of some point. It says nothing about density or accuracy.

Six points flung to the corners of a library floor reported '75%, good'. Your final survey reported a lower percentage purely because you chose a finer grid square, increasing the denominator. Point count and spacing are the real density figures. Reporting both is honest; reporting only the percentage would have been misleading.

4.4 Reading the export

A NetSpot .net.spu file is a zip containing a SQLite database. Unzipping it gives you the floor plan images and database.db3. Five tables matter: Zones (one per floor, with the scale factor), Snapshots (one survey run), ScanPoints (67 rows, with map coordinates), WlanNetworks (1,523 rows, joined to scan points), and Maps.

Every number in your report came from querying that database. None of it came from reading colours off a heatmap. That distinction is worth making explicitly — it is the difference between operating a tool and analysing data.

Part 5 — Self-test

Answer these without notes. Anything you stumble on, go back to the section it came from.

Physics and fundamentals

- Why is –60 dBm ten times worse than –50, not 'a bit worse'?
- Why is –67 the number people design to?
- What is SNR, and why is strong signal with poor SNR a different problem from weak signal?
- Why does your report contain no SNR finding?
- Why did holding the phone away from your body matter?

Structure and channels

- You saw 94 BSSIDs. How did you get to eight access points? What does the 8a prefix indicate?
- Why are only channels 1, 6 and 11 usable in 2.4 GHz?
- Explain the difference between co-channel and adjacent-channel interference. Which is worse and why?
- An 80 MHz channel with primary 149 — what spectrum does it occupy, and what does that collide with?
- What is DFS, and why does it leave UNII-2 underused?
- Why might 40 MHz be better than 80 MHz in this building?

Design and your findings

- What is the difference between coverage and capacity, and which did you find?
- Why would adding access points have made this network worse?
- How can reducing transmit power improve a network?
- How much power headroom does your data show, and how did you calculate it?
- What is a sticky client and how does excessive overlap encourage one?
- How did you demonstrate vertical propagation between floors, and what prompted you to look?
- Your link rate was 96 Mbps at –50 dBm on a 433 Mbps device. What does that tell you, and why does it matter that it was measured independently of the beacon survey?

Method and judgement

- What does 'map coverage 75%, good' actually measure?
- Why is F-04 labelled unverified, and what specifically would you need to confirm it?
- Name three things this survey cannot tell you and why.
- Why did you exclude the third floor, and why is that a strength rather than a gap?
- What would you do differently with a spectrum analyser and a multi-radio survey adapter?
- How would the results differ at 3pm on a Saturday?

The one to rehearse most. "Walk me through this project." Ninety seconds: what you set out to do, what you expected, what you actually found, why it was the opposite, and what you'd fix. The reversal is the story — lead with it.